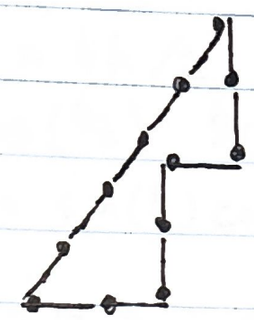
10/29/2026

14. Yes, multiple ways. The neatest I found was like this:



You can see it's formed after a 3-4-5 right triangle, which has area $(3\times4)/2 = 6$. So, if we "invert" the 3 matchsticks in the corner, we reduce the area by a rectangle of area $1\times2 = 2$, for a total area of $6-2 = \underline{4}$

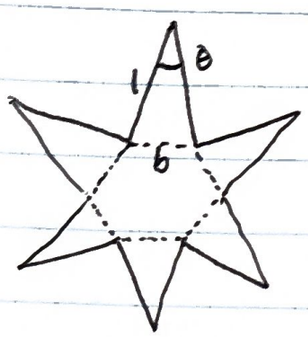
However, you can also achieve a shape with <u>any</u> area from 0 up to the area of the unit dodecagon (side-length) (which I believe is the maximum, $\approx 11.196$)
This can be shown with the fact you can make a 6-pointed star like so:



By varying the angle of the star's "points" you get a shape which continuously changes from one with an area as near to 0 as you desire, into the dodecahedron, achieving all areas continuously in between. In fact, the function which gives the angle needed for a desired area is:

$$f(A) = \arcsin\left(\frac{A-3\sqrt{3}}{6}\right) + \frac{\pi}{3}$$

I show this derivation on the next page.

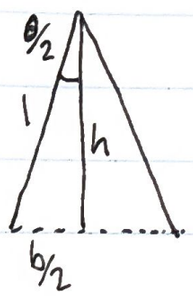



$$A_{star} = 6A_{spike} + A_{Hex}$$

$$A_{spike} = bh/2$$

$$h = \cos(\theta/2) \quad b/2 = \sin(\theta/2) \Rightarrow b = 2\sin(\theta/2)$$

$$A_{spike} = \sin(\theta/2)\cos(\theta/2)$$




$$A_{Hex} = \frac{3\sqrt{3}}{2}b^2 = \frac{3\sqrt{3}}{2}(2\sin(\theta/2))^2 = 6\sqrt{3}\sin^2(\theta/2)$$

$$A_{star} = 6\sin(\tfrac{\theta}{2})\cos(\tfrac{\theta}{2}) + 6\sqrt{3}\sin^2(\tfrac{\theta}{2})$$

$$A_{star}/6 = \sin(\tfrac{\theta}{2})\cos(\tfrac{\theta}{2}) + \sqrt{3}\sin^2(\tfrac{\theta}{2})$$

$$\frac{A_{star}}{6} = \frac{1}{2}\sin\theta + \sqrt{3}\left(\frac{1-\cos\theta}{2}\right) \quad \text{* double angle ident.}$$

$$\frac{A}{6} - \frac{\sqrt{3}}{2} = \frac{1}{2}\sin\theta * - \frac{\sqrt{3}}{2}\cos\theta$$

$$= R\sin(\theta+\phi)$$

$$= R\sin\theta\cos\phi + R\cos\theta\sin\phi \quad \text{* sum ident.}$$

$$= (R\cos\phi)\sin\theta + (R\sin\phi)\cos\theta$$

$$R\cos\phi = \frac{1}{2} \qquad R\sin\phi = -\sqrt{3}/2$$

$$\Rightarrow R^2(\cos^2\phi + \sin^2\phi) = \frac{1}{4} + \frac{3}{4}$$

$$R^2 = 1 \qquad R = 1$$

$$\Rightarrow \frac{\sin\phi}{\cos\phi} = \frac{-\sqrt{3}/2}{1/2} = -\sqrt{3}$$

$$\tan\phi = -\sqrt{3}$$

$$\phi = \arctan(-\sqrt{3}) = -\pi/3$$

$$\frac{A-3\sqrt{3}}{6} = \sin(\theta - \tfrac{\pi}{3})$$

$$\arcsin\left(\frac{A-3\sqrt{3}}{6}\right) = \theta - \pi/3$$

$$\underline{\arcsin\left(\frac{A-3\sqrt{3}}{6}\right) + \pi/3 = \theta}$$